

Pretraining Progress — 68.7M TransformerLM

research-v2 Pretraining Project (Kaggle GPU Kernels) | Date: 7-8-2026

Live Status

- Kernel: roronoazoro3008/research-v2-pretrain — running on 1x Tesla T4, 12h session
- Progress: step ~29,600 / 50,000 (started from step 24,500, auto-resumed)
- Throughput: ~25.9-27.4k tok/s at ~2.2 GB VRAM, ~2.5 s/step
- Val metrics @ step 29,500: loss 4.6422, perplexity 103.77, accuracy 25.86%
- Checkpoints auto-uploaded to Kaggle dataset every 500 steps
- Projection: session ends ~step 41k; one short follow-up run completes 50k

Model & Data

- Architecture: Dense decoder-only Transformer — RMSNorm, SwiGLU, RoPE, no bias/weight tying
- Size: 68.7M params (vocab 32k, d_model 576, 6 layers, 8 heads, d_ff 2304), 2048 context
- Optimizer: AdamW (lr 3e-4 to 3e-5, wd 0.1) with FP16 autocast + GradScaler
- Corpus: custom research/coding mixture (~1.28 GB, ~319M tokens), SentencePiece BPE 32k
- Budget: 50,000 steps x 2048 x 32 batch = ~3.28B tokens (~10 epochs)

Metrics Progression

Step	Val Loss	Val PPL	Val Acc
8,000	-	-	24.8%
24,500	4.6783	107.59	25.52%
29,500 (now)	4.6422	103.77	25.86%

Key History & Fixes

- 7 early kernel failures resolved (NaN losses, OOM, RoPE/DataParallel crashes, missing tokenization)
- Checkpoint/resume system built: uploads every 500 steps, disk-leak bug fixed, lazy Kaggle auth added
- New account required Persona identity verification to unlock GPU + internet + 30h/week quota

Roadmap

- Finish 50k steps with a short follow-up push
- Post-training: SFT + preference tuning (DPO/KTO-style) on the final checkpoint
- Run evaluation harness alongside training

Date: _____

Supervisor Signature: _____